

AMR Final Project

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Abstract—Write your abstract here.

Index Terms—Write up to three keywords about your work.

I. PROJECT DESCRIPTION

A. Motivation

//show what we have learned in the course Autonomous Mobile Robots and experience in realizing a realistic real life project.

Describe the context of your work and the motivation for it.

B. Problem Statement

//brief general task description

Describe the problem you are addressing in the work.

C. Proposed Approach

//which algorithms we chose to solve the task

Write a short summary of your proposed approach.

II. DESCRIBING THE CODE

Describe the code you have implemented and its functionality.

A. Task 1: Path and Motion Planning

B. Task 2: Localization

C. Task 3: Environment Exploration

III. EXECUTING THE CODE

This is a section in which you describe how you executed your code. Start with describing the environment and then describe how you executed the code in the simulation and on the real robot.

A. The Simulation

//dont forget to add pictures and videos of the simulation

B. The Real Robot

//dont forget to add pictures and videos of the real robot

IV. CHALLENGES

Describe the challenges you faced during the project and how you overcame them.

A. x

//challenges in the simulation

B. Connecting to the Robot

//challenges in connecting to the real robot

C. y

//challenges in the execution of the code on the real robot from translation from the simulation to the real world

D. Time Management

//challenges in time management and Rene didnt work at all on the project

V. CONCLUSIONS

Write a short summary of your conclusions and the lessons learned during the project.

STATEMENT OF ORIGINALITY

I, the undersigned below, declare that this work has not previously been submitted to this or any other university and that it is, unless otherwise stated, entirely my own work. The report was, in part, written with the help of the AI assistant Claude Code as described in the appendix. I am aware that content generated by AI systems is no substitute for careful scientific work, which is why all AI-generated content has been critically reviewed by me, and I take full responsibility for it.

Date

Signature

Signature

Signature

APPENDIX

how i used ai

*Submitted to the Department of Computer Science at Hochschule Bonn-Rhein-Sieg in partial fulfilment of the requirements for the degree of Master of Science in Autonomous Systems

†Submitted in September 2026